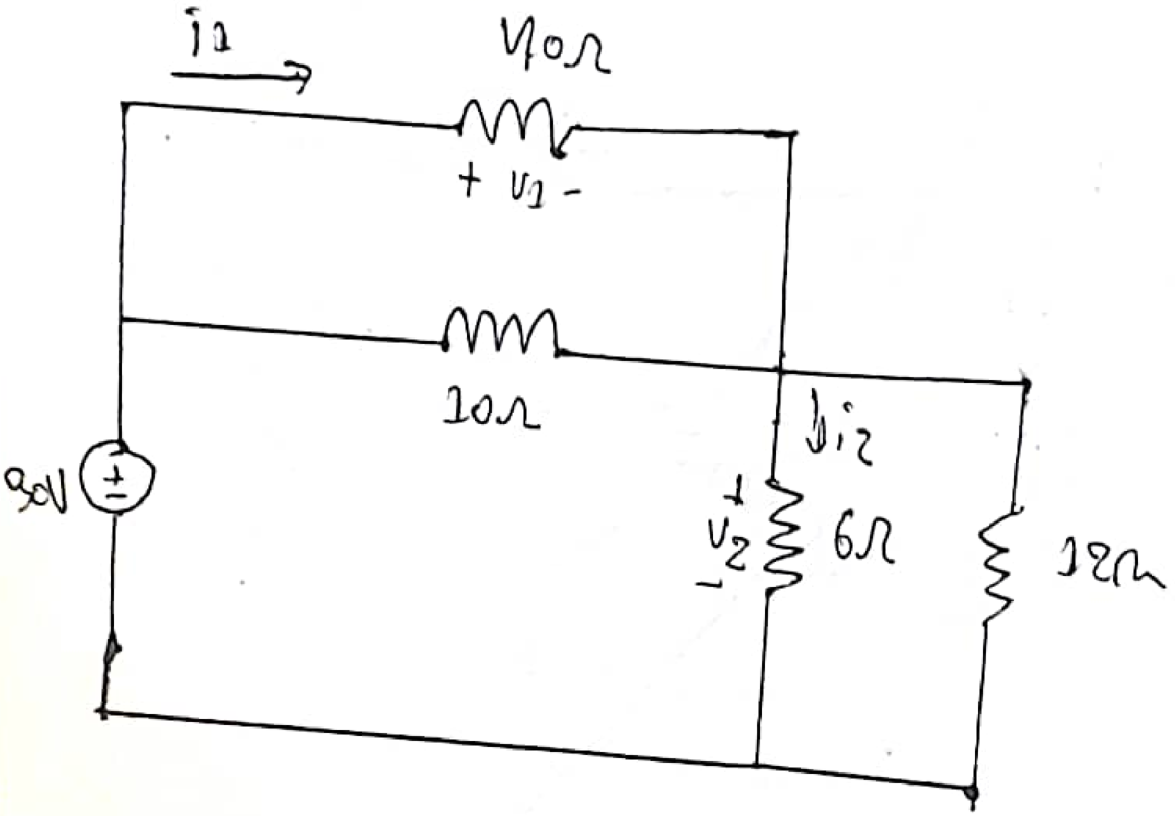
Oli Ahmed

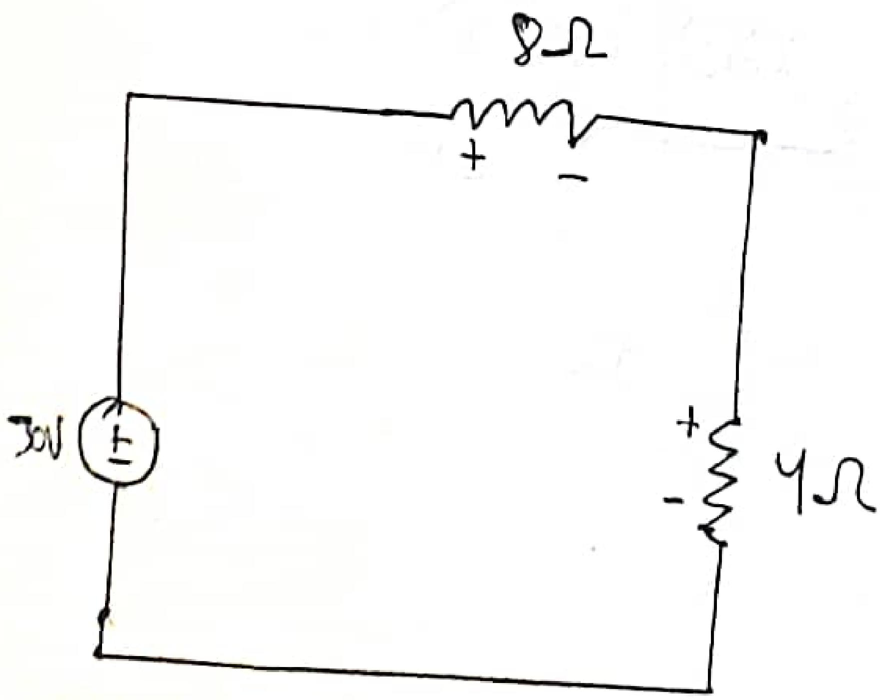
Id: 2019-3-60-073

Quiz 1, CSE-209

Ans to the ques no - 2



a)



using voltage divide rule,

$$V_1 = \frac{8}{4+8} \times 30V$$

$$= 20 V$$

and

$$V_2 = \frac{4}{4+8} \times 30V$$

$$= 10V$$

Ans.

b). we know,

$$V = IR$$

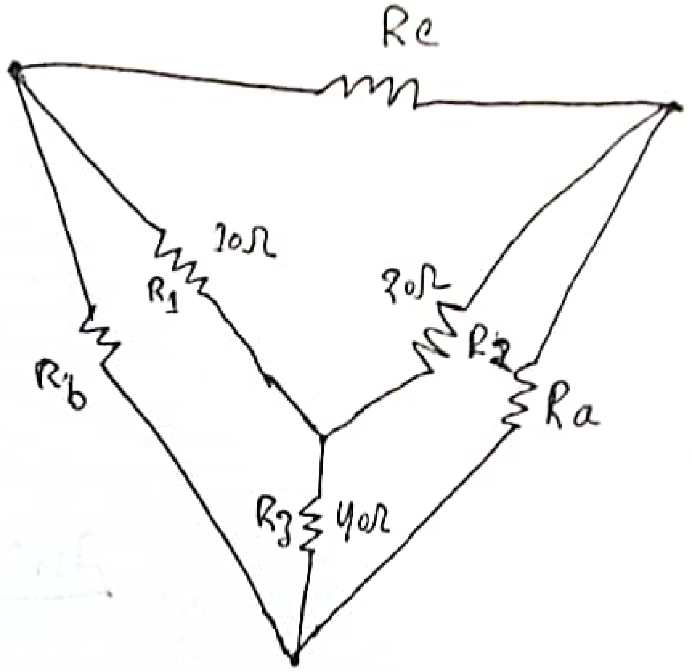
$$\therefore I = \frac{V}{R}$$

$$i_1 = \frac{20}{40} = 0.5 A$$

$$i_2 = \frac{10}{6} = 1.667 A$$

Ans.

Ans to the ques no-2



$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$= \frac{2400}{10} \Omega$$

$$= 240\Omega$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$= \frac{1400}{20}\ \Omega$$

$$= 70\ \Omega$$

$$R_C = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

$$= \frac{1400}{40}\ \Omega$$

$$= 35\ \Omega$$

Ans.